Social Network Data Analysis using AI

M. Ankitha (160123729017)
N. Archana (160123729018)

B.E AIML - Section J
Mentor: Mr. Panigrahi Srikanth

Academic Year: 2025-26

Introduction

- Online Social Networks (OSNs) enable content sharing and interaction
- Data is high-dimensional, unstructured, and multilingual
- Helps identify user behavior and patterns
- t-SNE visualizes high-dimensional data in 2D/3D

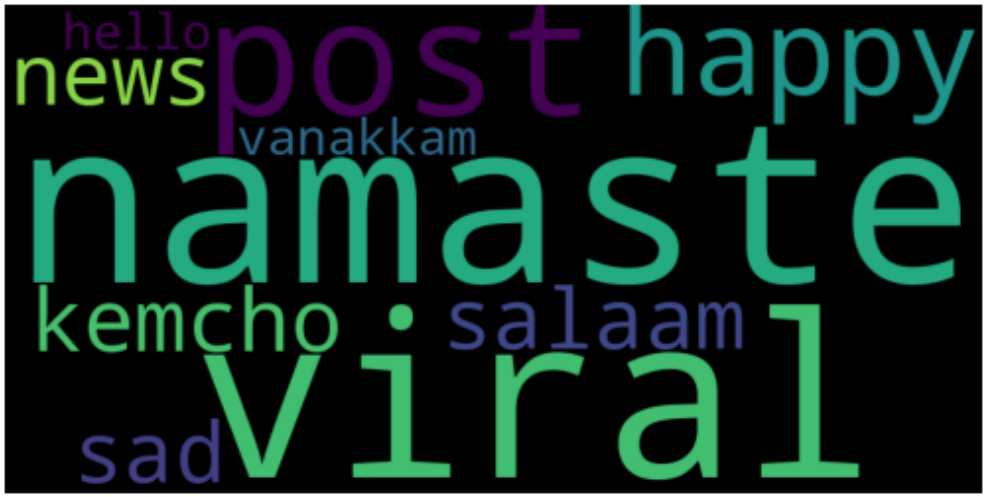
Dataset Description

- Synthetic Instagram dataset (5 lakh records)
- User data (ID, followers, location)
- Content data (captions, timestamps)
- Interaction data (likes, comments, shares)
- Network data (friends, communities)

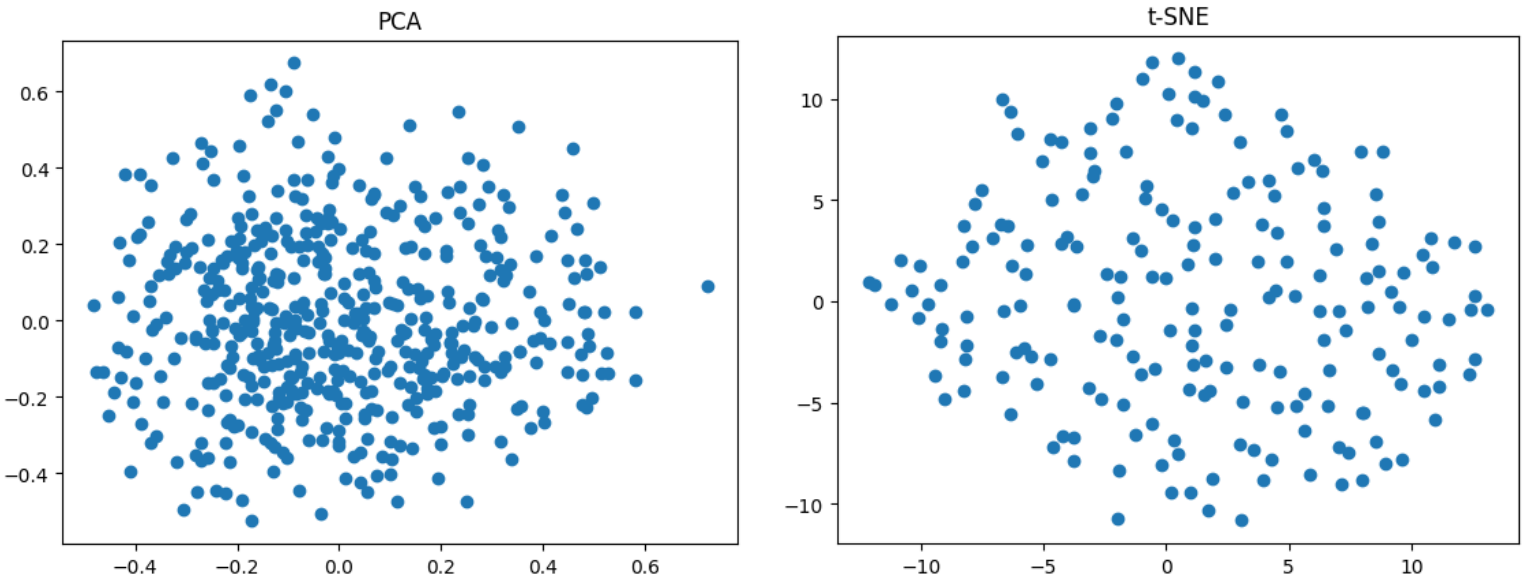
Pipeline

- Data Generation
- Text Preprocessing
- Sentiment Analysis
- TF-IDF Vectorization
- Cosine Similarity
- Graph Construction
- PCA + t-SNE
- KMeans Clustering
- BERT Embeddings
- Dashboard Visualization

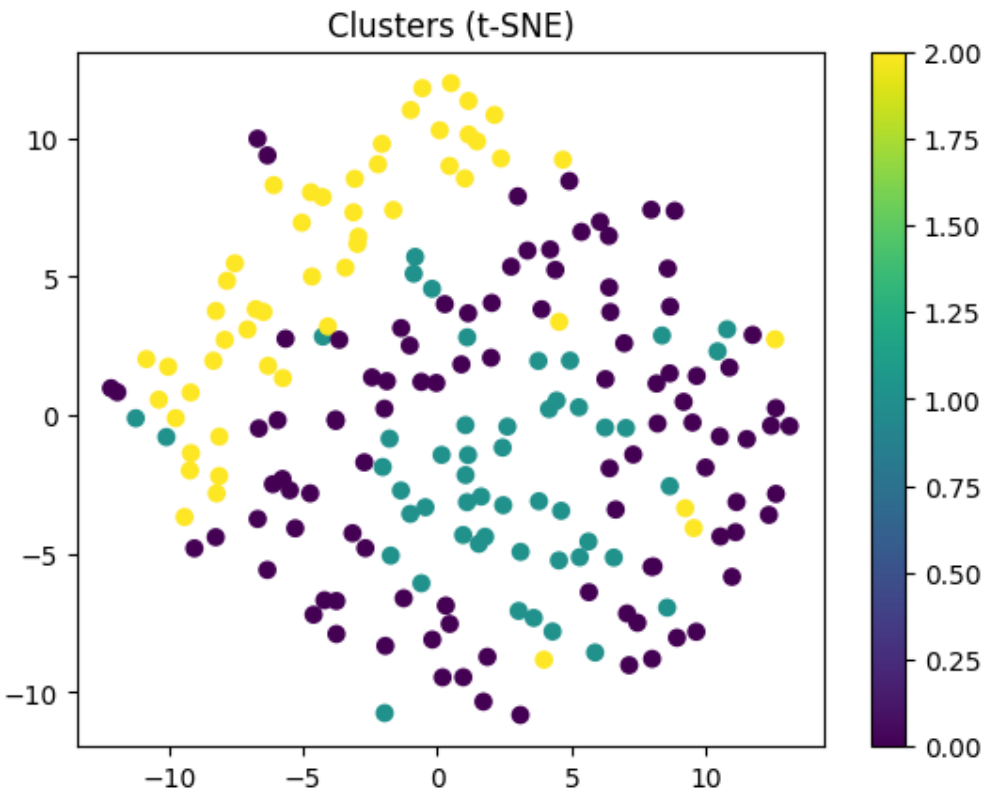
Word Cloud


hello
news
post
happy
vanakkam
namaste
kemcho
salaam
sad
viral

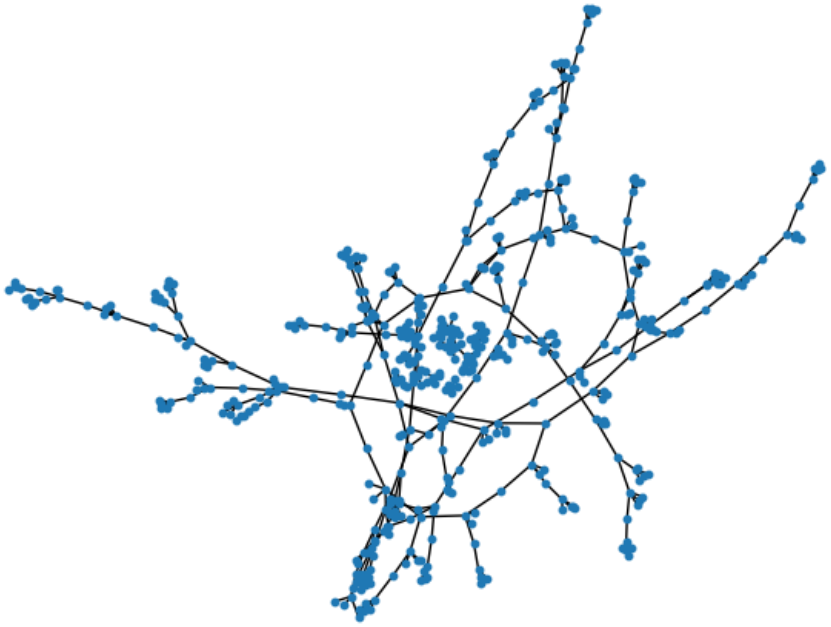
PCA and t-SNE



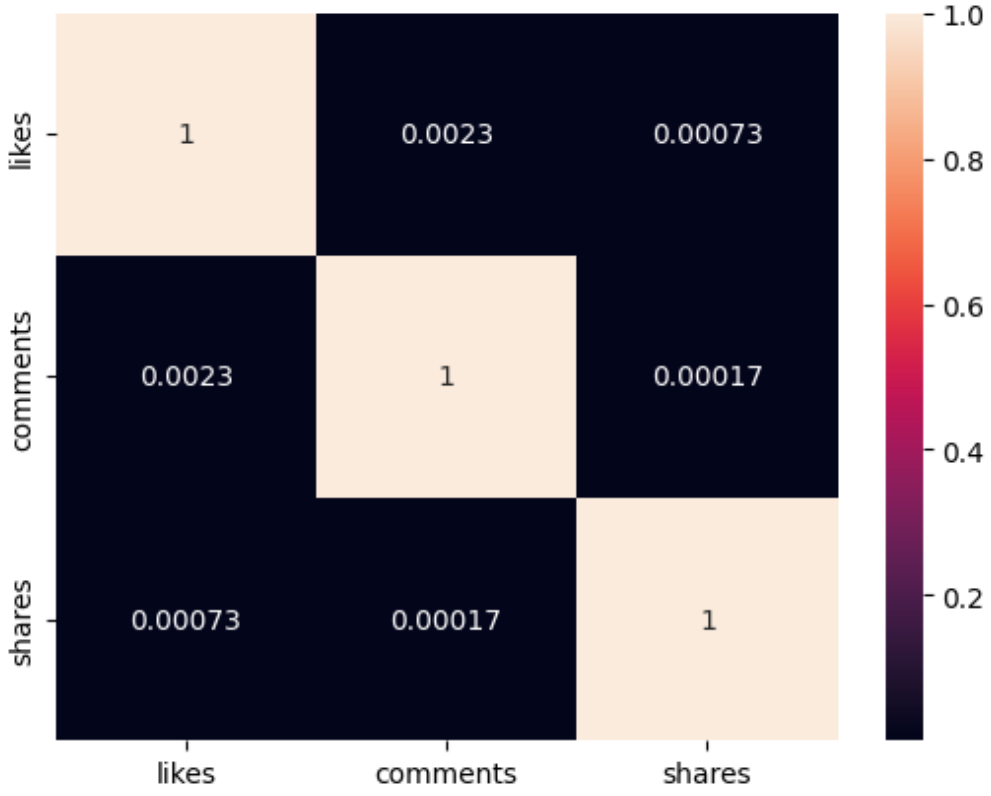
Clustering


Clusters (t-SNE)
10
5
0
−5
−10
−10
−5
0
5
10
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00

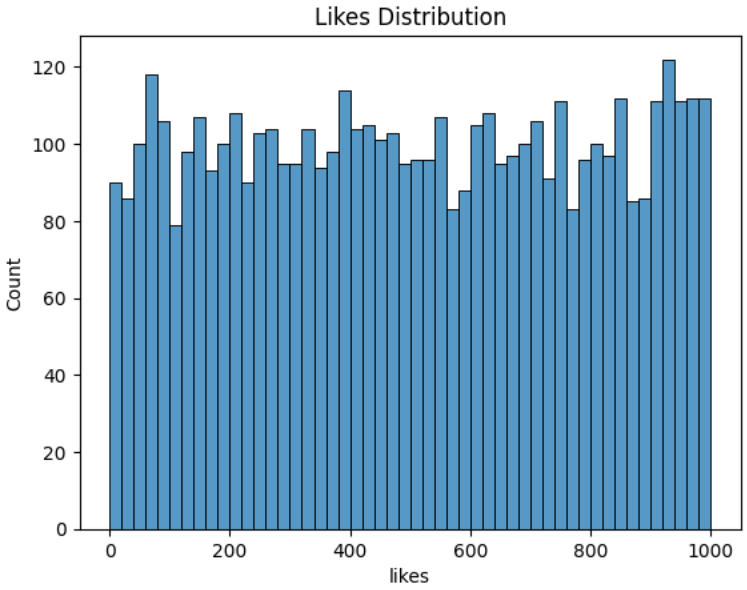
Network Graph

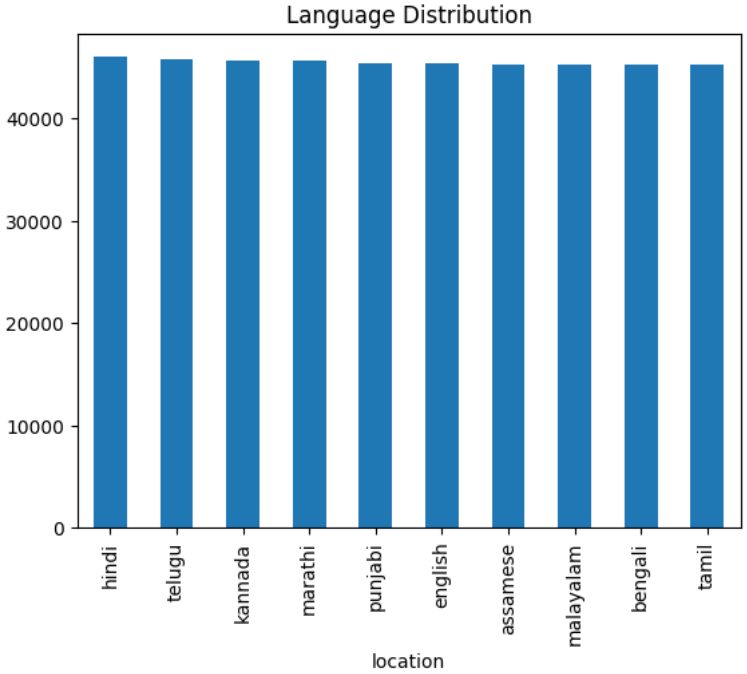


Heatmap


likes
comments
shares
1
0.0023
0.00073
0.0023
1
0.00017
0.00073
0.00017
1
likes
comments
shares
1.0
0.8
0.6
0.4
0.2

Distribution Graphs





Interactive Dashboard

Conclusion

- Successfully analyzed large-scale social media data
- Extracted insights from multilingual dataset
- Visualized data using PCA and t-SNE
- Identified patterns using clustering
- Built interactive dashboard for insights

Thank You